SOFTWARE REQUIREMENTS SPECIFICATION

TRUSTCHARGE

AI-Based Centralized EV Charging Reliability and Smart Reservation Platform

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification is to define the requirements of TrustCharge, an AI-Based Centralized EV Charging Reliability and Smart Reservation Platform. The document specifies the intended functions, interfaces, architecture, data requirements, security requirements, constraints, and evaluation criteria for the system. It provides a common technical reference for the four-member development team and for project review and testing.

1.2 Scope

The proposed system is a centralized platform that unifies fragmented Electric Vehicle (EV) charging networks into a single, seamless ecosystem. It integrates data and services from multiple independent charging providers so that users interact with one application instead of several. Using AI models, the system predicts charger health, assesses functional reliability, and estimates queue waiting times, and it supports cross-network slot reservation through a unified membership and wallet system. The core objective is to eliminate range anxiety and streamline the EV charging experience for individual users, fleet owners, and station operators.

1.3 Intended Audience

This document is intended for the project team, project guide, evaluators, testers, and future maintainers. It can also be used as the baseline for database design, API design, AI model integration, integration testing, and project demonstration.

1.4 Definitions and Abbreviations

EV – Electric Vehicle; AI – Artificial Intelligence; ML – Machine Learning; API – Application Programming Interface; SRS – Software Requirements Specification; UI – User Interface; TLS/SSL – secure transport protocols; ACID – Atomicity, Consistency, Isolation and Durability; OCM – OpenChargeMap; OSM – OpenStreetMap; ETA – Estimated Time of Arrival.

2 Overall Description

2.1 Product Perspective

TrustCharge follows a client-server architecture with a React.js frontend styled using Tailwind CSS, a Node.js/Express backend exposing REST APIs, a MySQL relational database, Firebase Authentication for identity management, and an AI layer responsible for charger health prediction and queue waiting time estimation. Station and location data are sourced from the OpenChargeMap and OpenStreetMap APIs and merged with real-time provider data within the backend.

2.2 Product Functions

Major functions include secure user registration and unified membership management, centralized discovery of charging stations across multiple providers, AI-driven charger reliability scoring and queue prediction, cross-network prepaid slot reservation, real-time turn-by-turn navigation, a unified digital wallet, and membership-based priority access during peak demand.

2.3 User Classes

The primary users are individual EV owners and fleet owners seeking reliable charging stations and predictable waiting times. A secondary user class consists of charging station operators, who benefit from increased station utilization and visibility. An administrative role may be included for managing provider integrations, monitoring AI model performance, and maintaining supported data sources.

2.4 Operating Environment

The application is intended to run as a responsive web application accessible on desktop and mobile-sized screens through modern browsers. The backend runs on a server environment capable of hosting REST APIs and AI inference services, MySQL stores persistent relational data, and the system depends on continuous internet connectivity for real-time station data, navigation, and payment operations.

2.5 Assumptions and Dependencies

The system assumes that integrated charging-provider APIs and the OpenChargeMap/OpenStreetMap services remain available and reasonably accurate. The quality of AI-based reliability scoring and queue prediction depends on the availability and quality of historical usage and station-status data. Payment and wallet operations depend on the availability of a supported payment gateway.

3 System Analysis

3.1 Existing System

Existing applications such as Tata Power EZ Charge, Statiq, ChargeZone, and Bolt.Earth provide basic functionality including locating nearby charging stations on a map, basic navigation to reach a station, and in-app payment for charging sessions. However, each application operates independently and serves only its own provider network.

3.2 Problems in Existing System

- Separate applications are required for different charging providers.
- No centralized membership or wallet system across networks.
- Complete absence of AI-based reliability prediction, so users often arrive at broken or unavailable chargers.
- Limited or no queue-waiting-time prediction capability.
- Users cannot easily compare station availability and reliability across different networks.

3.3 Problem Statement

To address the unreliability of fragmented EV charging networks, there is a critical need for a platform that centralizes real-time availability data from different providers and integrates a unified reservation system, thereby eliminating long waiting times and severe range anxiety.

4 Proposed System

4.1 Proposed Solution

TrustCharge introduces a centralized hub that connects multiple charging providers into a single application using AI and real-time data. The platform combines an AI Reliability Score and charger health prediction, a unified membership wallet spanning all networks, a Cross-Net Reserve capability for booking slots in advance across providers, and an AI Queue Predict feature for accurate waiting-time estimation.

4.2 Advantages

- A single application experience across previously fragmented charging networks.
- Reduced range anxiety through AI-verified charger reliability before arrival.
- Predictable waiting times through AI-based queue estimation.
- Simplified payments through one unified wallet and membership.
- Higher utilization and satisfaction for participating station operators.

4.3 System Workflow

A typical workflow begins when a user searches for or is recommended a charging station based on current location and battery status. The backend aggregates station data from OpenChargeMap, OpenStreetMap, and integrated provider feeds, while the AI layer scores charger reliability and estimates queue waiting time for each candidate station. The user selects a station, optionally reserves a slot in advance through Cross-Net Reserve, and is guided to the station through real-time navigation. Payment for the session is settled through the unified wallet.

5 System Architecture

5.1 Architecture Components

The architecture contains four major layers: the presentation layer (React.js frontend styled with Tailwind CSS), the application layer (Node.js/Express backend exposing REST APIs), the data layer (MySQL relational database), and the AI and integration layer (Firebase Authentication, AI models for charger health and queue prediction, and the OpenChargeMap/OpenStreetMap external data services).

5.2 Data Aggregation Layer

The data aggregation layer is responsible for pulling station location, status, and connector data from OpenChargeMap and OpenStreetMap and normalizing it alongside data from integrated charging-provider networks so that the rest of the system can treat all stations through one consistent interface.

5.3 AI Prediction Layer

The AI prediction layer hosts two core models: a charger health/reliability model that predicts whether a given charger is likely to be functional at the time of arrival, and a queue-prediction

model that estimates waiting time at a station based on historical usage patterns and current occupancy signals.

5.4 Reservation and Wallet Layer

This layer manages prepaid slot bookings across providers (Cross-Net Reserve), maintains the unified wallet balance and transaction history, and enforces membership-based priority access rules during periods of peak demand.

6 Detailed Module Description

6.1 User Account & Membership Module

Handles secure registration and authentication through Firebase Authentication, and maintains a unified membership profile that is recognized across all supported charging networks.

6.2 Centralized Station Discovery Module

Integrates data from OpenChargeMap and OpenStreetMap, along with connected provider feeds, to present comprehensive real-time station information such as location, connector type, and availability within one unified interface.

6.3 AI Reliability & Queue Prediction Module

Analyzes historical and real-time station data to produce an AI Reliability Score for each charger and to predict queue waiting time, helping users avoid non-functional chargers and long queues.

6.4 Smart Reservation Module (Cross-Net Reserve)

Allows users to perform prepaid slot bookings and manage reservations across different, independently operated charging networks from within a single interface.

6.5 Navigation & Routing Module

Provides turn-by-turn navigation to a recommended charging station based on the user's current location and current battery status.

6.6 Unified Wallet & Priority Access Module

Maintains a single membership-linked wallet usable across all networks and facilitates membership-based priority access so that eligible users can secure charging slots during peak demand periods.

7 Functional Requirements

ID	Requirement
FR-01	The system shall provide secure user authentication and manage a unified membership account accessible across all supported charging networks.
FR-02	The system shall integrate data from OpenChargeMap and OpenStreetMap to display comprehensive, real-time charging station information from multiple providers in one unified interface.

ID	Requirement
FR-03	The system shall utilize AI models to analyze historical and real-time data to predict individual charger health/reliability.
FR-04	The system shall provide accurate queue waiting time estimations for a selected charging station.
FR-05	The system shall allow users to perform prepaid slot bookings across different charging networks (Cross-Net Reserve).
FR-06	The system shall allow users to view and manage their active and past reservations.
FR-07	The system shall provide turn-by-turn navigation to recommended charging stations based on the user's current location and battery status.
FR-08	The system shall facilitate membership-based priority access to secure charging slots during peak demand.
FR-09	The system shall maintain a unified digital wallet supporting payment across all integrated networks.
FR-10	The system shall allow users to compare available stations across networks based on reliability score, distance, and estimated waiting time.

8 Non-Functional Requirements

ID	Requirement
NFR-01 Scalability	The platform must be highly scalable to seamlessly accommodate new charging providers and expanding geographic regions without performance degradation.
NFR-02 Performance & Latency	AI predictions, route recommendations, and real-time station availability updates must be processed and delivered to the user rapidly with minimal latency.
NFR-03 Security	The system shall implement robust security protocols and encryption for the unified wallet system, secure reservations, and end-to-end protection of user data.
NFR-04 Usability	The interface shall present station reliability, availability, and waiting-time information in a clear and easily understandable format for users without technical expertise.
NFR-05 Reliability	The system shall degrade gracefully when an external provider, OpenChargeMap, or OpenStreetMap API is temporarily unavailable.
NFR-06 Maintainability	Frontend, backend, and AI modules shall remain modular so individual components can be upgraded independently.

9 External Interface Requirements

9.1 User Interface

The React.js frontend, styled with Tailwind CSS, shall provide a responsive interface for account management, station discovery, reservation booking, navigation, wallet management, and viewing AI-generated reliability and queue predictions across desktop and mobile-sized displays.

9.2 Backend Interface

The Node.js/Express backend shall expose secure REST endpoints for authentication, station data, reservations, wallet transactions, and AI prediction results.

9.3 External Data Interfaces

The system shall integrate the OpenChargeMap and OpenStreetMap APIs for station location and navigation data, Firebase Authentication for identity management, and a supported payment gateway for wallet top-ups and transactions.

10 Data Requirements

The system shall maintain structured data for users, charging stations, individual chargers, reservations, wallet transactions, AI reliability scores, and queue predictions. Station and connector data sourced from OpenChargeMap and OpenStreetMap shall be kept synchronized with provider feeds. Historical charger usage and status data shall be retained to support ongoing training and refinement of the AI reliability and queue-prediction models.

11 Database Requirements

Entity	Key Fields	Purpose
User	user_id, name, email, password_hash, membership_tier	Authentication and unified membership management
Station	station_id, provider, name, latitude, longitude, address	Charging station location and provider details
Charger	charger_id, station_id, connector_type, status, reliability_score	Individual charger status and AI reliability score
Reservation	reservation_id, user_id, charger_id, slot_time, status	Cross-network slot booking records
Wallet	wallet_id, user_id, balance, updated_at	Unified wallet balance per user
Transaction	transaction_id, wallet_id, amount, type, created_at	Wallet payment and top-up history
QueuePrediction	prediction_id, station_id, predicted_wait, generated_at	AI-generated queue waiting time estimates

12 Use Case Analysis

ID	Requirement
UC-01	User Registration and Login – User creates an account and authenticates securely.
UC-02	Station Discovery – User searches for nearby charging stations across all integrated networks.
UC-03	Check Charger Reliability – User views the AI reliability score before travelling to a station.

ID	Requirement
UC-04	View Queue Estimate – User views the predicted waiting time at a selected station.
UC-05	Reserve Slot – User books a prepaid charging slot in advance through Cross-Net Reserve.
UC-06	Navigate to Station – User receives turn-by-turn navigation to the selected station.
UC-07	Wallet Payment – User pays for a charging session through the unified wallet.
UC-08	Priority Access – Eligible member secures a charging slot during peak demand.

13 Hardware and Software Requirements

ID	Requirement
Frontend	React.js, Tailwind CSS, modern browser
Backend	Node.js, Express for API endpoint management
Database	MySQL for relational database storage
Authentication	Firebase Authentication for secure identity management
AI Models	AI-based charger health prediction and AI-based queue waiting time estimation
External APIs	OpenChargeMap and OpenStreetMap APIs for location/navigation data
Development Hardware	Standard modern processor (e.g., Intel Core i3 or equivalent), 4 GB RAM minimum, 50 GB available disk space, standard internet connection for API access

14 Security and Privacy Requirements

- User passwords and credentials shall be managed through Firebase Authentication using secure, industry-standard practices.
- All frontend-backend communication shall use TLS/SSL in deployment environments.
- Wallet balances and payment transactions shall be encrypted in transit and at rest.
- API keys and third-party service credentials shall be stored in secure environment variables or a secret-management mechanism.
- Access to reservation and wallet records shall be restricted by authenticated user identity and authorization rules.

15 System Constraints

The system is limited by the coverage and update frequency of OpenChargeMap, OpenStreetMap, and integrated provider APIs; by the availability and quality of historical data used to train the AI reliability and queue-prediction models; and by the willingness of individual charging networks to share real-time status data. The project must also remain feasible within the time and resources available for a B.Tech final-year group implementation. The system shall not

guarantee charger availability, as this depends on third-party provider data accuracy.

16 Testing and Evaluation Requirements

Each module shall be unit-tested independently before integration. The AI reliability-scoring and queue-prediction models shall be evaluated using task-appropriate metrics such as accuracy, precision, recall, and mean absolute error for waiting-time predictions. The reservation and wallet modules shall be validated through deterministic transaction and booking-consistency tests. The complete system shall be tested for latency, external API failure handling, security, data integrity, and usability.

17 Future Scope

Area	Description
Global Expansion	Scaling the platform's footprint across multiple states and integrating international charging networks.
V2G Integration	Incorporating smart grid features to allow EV owners to sell energy back to the grid during peak hours.
Autonomous Fleets	Developing API endpoints for self-driving vehicle fleets to autonomously route to and reserve charging slots.
Dynamic Pricing	AI-driven cost recommendations helping users schedule charging sessions when grid rates are lowest.

18 Conclusion

TrustCharge is designed as a centralized, AI-powered EV charging platform that improves reliability and significantly reduces waiting times by unifying fragmented charging networks into a single ecosystem. By combining AI-based charger health prediction, queue estimation, cross-network reservation, and a unified membership wallet, the system offers a seamless charging experience while supporting station operators and encouraging wider EV adoption. This SRS establishes the baseline requirements for implementing, integrating, and evaluating the proposed B.Tech project while preserving a modular architecture for future expansion.

19 References

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